Haocheng Yin

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Education

Georgia Institute of Technology — **Atlanta, GA**
Ph.D. in Robotics — *August 2025 – Present*
Advisors: Prof. Lu Gan and Prof. Yongsheng Chen — GPA: 4.00/4.00

ETH Zürich — **Zürich, Switzerland**
M.S. in Computer Science — *September 2021 – October 2024*
Major: Machine Intelligence — GPA: 5.27/6.00

University of Illinois Urbana-Champaign — **Champaign, IL**
B.S. in Electrical Engineering — *September 2017 – May 2021*
ZJU-UIUC Dual Degree Program — GPA: 3.94/4.00

Zhejiang University — **Hangzhou, China**
B.E. in Electrical Engineering & Automation — *September 2017 – June 2021*
ZJU-UIUC Dual Degree Program — GPA: 3.96/4.00

Publications

(∗ indicates equal contribution)

[1] **Haocheng Yin***, Shuohan Tao*, Yongsheng Chen, and Lu Gan. "PhysCoRe: Physics-Corrected Residual World Models for Material-Aware Deformable Dynamics". In: *Conference on Robot Learning (CoRL)*. Proceedings of Machine Learning Research. PMLR, 2026. arXiv: 2607.20653 [cs.RO].

[2] Han Qi, **Haocheng Yin**, Aris Zhu, Yilun Du, and Heng Yang. "Inference-Time Enhancement of Generative Robot Policies via Predictive World Modeling". In: *IEEE Robotics and Automation Letters (RA-L)* (2026). DOI: 10.1109/LRA.2026.3673995. arXiv: 2502.00622 [cs.RO].

[3] Han Qi*, **Haocheng Yin***, and Heng Yang. "Control-oriented Clustering of Visual Latent Representation". In: *International Conference on Learning Representations (ICLR)*. 2025. arXiv: 2410.05063 [cs.LG].

Research Experience

Lunar Lab | Advisors: Prof. Lu Gan and Prof. Yongsheng Chen — **Georgia Tech**
Graduate Research Assistant — *August 2025 – Present*
Physics-Grounded World Models for Deformable Object Manipulation [1]

Computational Robotics Lab | Advisor: Prof. Heng Yang — **Harvard University**
Visiting Researcher — *March 2024 – February 2025*
Latent Representation and Generative Predictive Control for Visuomotor Policies [2, 3]

Soft Robotics Lab | Advisor: Prof. Robert Katzschmann — **ETH Zürich**
Project: *Hand Pose Capture and Retarget for Dexterous Manipulation* — *December 2023*

Optimization & Decision Intelligence Lab | Advisor: Prof. Niao He — **ETH Zürich**
Projects: *Bio-plausible Meta RL*; *Inverse RL from Suboptimal Demonstrations* — *September 2022*

Awards & Honors

CoRL 2026 Travel Award — **CoRL**
Awarded by the CoRL 2026 organizing committee — *October 2026*

College of Engineering (CoE) Conference Matching Fund — **Georgia Tech**
Supplemental conference travel support for CoRL 2026 — *September 2026*

Student Government Association (SGA) Conference Fund — **Georgia Tech**
Travel support for presenting research at CoRL 2026 — *September 2026*

Swiss-European Mobility Programme (SEMP) Scholarship — **ETH Zürich**
Funded research visit to Harvard University — *February 2024*

High Honors at Graduation — **UIUC**
Awarded for graduating with a GPA above 3.80 — *May 2021*

Dean's List in ECE Department — **UIUC**
Top-3 GPA in the ECE class for four consecutive years — *May 2021*

Undergraduate Technology Innovation Award — **Zhejiang, China**
Top 7% of student research projects across all universities in Zhejiang, China — *August 2020*

Presentations

NVIDIA Robotics Day — **Georgia Tech**
Presented PhysCoRe, physics-corrected world models for deformable object dynamics [1] — *October 2026*

Teaching Experience

ECE 365: Data Science and Engineering — **UIUC**
Teaching Assistant (remote) — *Spring 2021*

ECE 385: Digital Systems Laboratory — **Zhejiang University**
Teaching Assistant — *Fall 2020*

Academic Service

Reviewer

- IEEE International Conference on Robotics and Automation (ICRA) — *2027*
- Conference on Robot Learning (CoRL) — *2026*
- ICLR Workshop on World Models: Understanding, Modelling and Scaling — *2025, 2026*

Skills

Programming Languages
Python, C/C++, C#, MATLAB, JavaScript, SQL, HTML

Tools & Frameworks
PyTorch, NVIDIA Isaac Sim, NVIDIA Isaac Lab, NVIDIA Warp, NVIDIA Newton, MuJoCo, ROS 2